Mamba Cross-Modal Information Fusion Self-Distillation Model for Joint Classification of LiDAR and Hyperspectral Data

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***Abstract*—Recent studies have found that compared to single-modal data, the joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) multimodal data can use their complementary information to further improve the accuracy of land-cover classification. However, due to the significant differences between multimodal data, the complementarity among them is difficult to be fully exploited and used, and the features after fusion are not refined and optimized, which limits the further improvement of land-cover classification accuracy. To alleviate these issues, a novel Mamba cross-modal information fusion self-distillation (Mb-CMIFSD) model is designed. Specifically, Mb-CMIFSD first uses conventional convolutional neural networks (CNNs) to transform each patch into a token sequence. Second, a Mamba cross-modal information fusion (MCMIF) module is developed to combine cross-modal attention (CMA) with bidirectional Mamba mechanism, which can better explore the complementarity of multimodal remote sensing (RS) data and obtain more discriminative multimodal fusion features (FFs). Finally, a prototype constrained self-distillation (PCSD) module is designed to use the constructed prototype orthogonal regularization (POR) knowledge distillation function to further refine cross-modal FFs, thereby enhancing the robustness and adaptability of feature extraction. The experimental results on three benchmark HSI and LiDAR datasets show that the designed Mb-CMIFSD model has higher classification accuracy compared to other state-of-the-art methods, and the ablation experiments also confirm the positive effect of the designed two key modules.**

***Index Terms*—Mamba cross-modal feature fusion, multimodal data classification, prototype constrained self-distillation (PCSD).**

I. Introduction

IN RECENT years, with the rapid development of satellite and communication technology, remote sensing (RS) data has been widely used for monitoring and analyzing the Earth's surface, including land-cover classification, environmental monitoring, urban planning, and disaster monitoring [1]. For example, hyperspectral imaging (HSI), as an important means of RS data acquisition, is widely used in land-cover classification tasks based on RS data because it can simultaneously obtain rich spectral and spatial information of the Earth's surface [2]. However, it has been found that relying only on HSI single-modal RS data greatly hinders the improvement of the classification accuracy of HSI images because it cannot well distinguish land covers with similar spectral features [3]. Therefore, using other sensors to collect other RS data of complementary modalities of HSI for the same scene, i.e., using the complementarity between multimodal RS data, and investigating the joint classification method of multimodal RS data has become an emerging and popular research topic in the field of computer vision today [4].

In the multimodal RS data classification task, since light detection and ranging (LiDAR) data can provide elevation information, which is an effective supplement to the spectral information of HSI data, so a lot of multimodal RS classification methods based on HSI and LiDAR data have been proposed. Such as Pedergnana et al. [5] calculated extended morphological attribute profiles of HSI and LiDAR data for multimodal RS collaborative classification. Khodadadzadeh et al. [6] used morphological methods to extract fused features from HSI and LiDAR data for multimodal RS joint classification. In recent years, with the rapid development of deep learning, the fusion classification technology of HSI and LiDAR data based on convolutional neural networks (CNNs) has received increasing attention. For example, Hong et al. [7] designed a new multisource RS data (i.e., HSI and LiDAR) classification technology, which can effectively extract compact feature representations of multimodal RS data. Hong et al. [8] first proposed a universal and unified multimodal deep-learning RS image classification framework, providing an effective solution for multisource and multimodal RS image classification. Zhong et al. [9] designed a spectral spatial residual network (SSRN) that can better use spectral and spatial information to improve classification accuracy. Roy et al. [10], [11] modeled the spectral and spatial features extracted by squeeze and excitation ResNet, explored lightweight paradigms, and combined them with bag of features learning to achieve accurate land-cover classification.

Although CNNs have shown outstanding performance in multimodal RS data classification, they are unable to capture sequence attributes, especially long-term dependencies of sequences. While a transformer is very effective in

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processing and analyzing sequence data due to its self-attention technique [12]. Therefore, many transformer-based multimodal RS data classification methods have been proposed, e.g., Hong et al. [13], by introducing SpectralFormer, which learns the spectral information using neighboring bands and builds a cross-layer transformer encoder module. Feng et al. [4] proposed a cross-modal self-supervised contrast learning approach, i.e., intramodal and intermodal contrast learning, which enables the integration of cross-attention and convolution, and further integrates heterogeneous fea tures more efficiently to produce a powerful fused feature to improve the classification accuracy. Roy et al. [3] devised a new multihead cross-patch attention mechanism to fuse complementary modalities and HSI patch tokens for land-cover classification. Song et al. [14] proposed a bidirectional interactive fusion network for HSI and LiDAR co-classification, where the interaction of information is achieved by mutual induction between the two modalities. Compared to the pure CNN framework, although the classification accuracy of multimodal RS data is improved by introducing the transformer, its attention computation comes at the cost of squared complexity, and has some limitations in terms of modeling efficiency and memory usage as the length of the input sequence increases. Recently, since the state space model (SSM) [15] can not only establish long-distance dependencies through state transitions, but also has near-linear computational complexity, especially Mamba [16] by merging time-varying parameters into SSM with hardware optimization, then the Mamba-based multimodal RS data classification method has also been proposed, and achieves a more efficient performance than the transformer for more efficient performance [17].

Based on the analysis of the existing methods mentioned above, it can be found that although various CNN-based or transformer-based methods have their own advantages, there are still some challenging issues worth further exploration and improvement, i.e.,

1) The complementarity between modalities has not been fully explored and used. Although multimodal RS data are imaged in the same scene, each modality has its own unique advantages due to the differences in imaging mechanisms, so a simple cross-modal feature fusion strategy makes it difficult to fully exploit the complementary between heterogeneous features from different modalities, due to the fact that: ① multimodal raw features from independent feature extraction branches with semantic misalignment between them; ② dependencies between pixels within a patch are poorly modeled, thereby affecting the effectiveness of perceiving global contextual information; and ③ the importance of different modalities varies, and improper fusion ratios can lead to underuse of information in weaker modalities.
2) No dedicated mechanism designed to further refine the fusion feature (FF). Although the cross-modal FF effectively uses the complementarity between multimodal RS data, if it is not further refined, it will inevitably affect the generalization ability of the model. This is because in the joint classification task of multimodal RS data, even if different pixels have the same semantic, their corresponding "patches" may show different visual characteristics due to the influence of neighboring pixels, while "patches" with similar visual characteristics may belong to different semantic categories, i.e., there may be a semantic gap problem of "large intraclass differences and small interclass differences." Therefore, refining FFs to enhance the robustness of model feature extraction is also a key issue worth studying in multimodal RS data joint classification tasks.

To address the challenging issues above, this article designs a novel Mamba cross-modal information fusion self-distillation (Mb-CMIFSD) model, which aims to combine the advantages of Mamba's global context awareness with knowledge distillation's feature refinement, in order to fully leverage the complementarities between multimodal RS data and obtain more robust FF, so as to further improve the accuracy of land-cover classification based on multimodal RS data. In summary, the main contributions of this article are reflected in the following three aspects.

1) In the designed Mb-CMIFSD model, a novel Mamba cross-modal information fusion (MCMIF) module is innovatively designed. This module integrates cross-modal attention (CMA) with the bidirectional Mamba mechanism, aiming to use Mamba's excellent long-range modeling capability to fuse the raw features from two modalities, in order to obtain more discriminative FFs by fully exploiting the complementarity of multimodal RS data.
2) In the Mb-CMIFSD model, a novel prototype constrained self-distillation (PCSD) module is innovatively designed. This module integrates prototype learning with knowledge distillation, aiming to use the constructed prototype orthogonal regularization (POR) Kullback-Leibler (KL) self-distillation function to further refine the cross-modal FFs, thereby enhancing the robustness and adaptability of feature extraction.
3) Based on three benchmark multimodal RS datasets, i.e., Houston, MUUFL, and Trento, we conducted extensive comparative experiments, and the designed Mb-CMIFSD model has higher classification accuracy than other state-of-the-art multimodal classification methods. The ablation experiment also confirmed the effectiveness of the designed two key modules, e.g., the MCMIF module improves the classification accuracy of the Trento dataset by 4.29% compared to the baseline, and then by introducing the PCSD module, the classification accuracy can be further improved by 8.85%.

II. Related Work

A. Land-Cover Classification of RS Data

Land-cover classification [18] is an important research topic for RS data analysis, which can help identify different land use types, and thus, provide data support for land use planning and management. First, HSI, as a typical representative of RS data, has been proven to be valuable in land-cover classification due to its rich spectral and spatial information [19], so many classification methods for HSI have been proposed, e.g., Feng et al. [20] proposed a pseudo label-assisted subdomain adaptation

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(PASDA) method for cross-domain HSI classification problems, which improves HSI classification accuracy by jointly considering the subdomain structures of the source and target domains. Wu et al. [21] designed a multitask multiobjective evolutionary network for the joint visual perception task of hyperspectral pansharpening and hyperspectral classification in HSI. Ma et al. [22] designed a spatial pooling transformer network and a noise-tolerant learning method for the classification of noisy HSI. In addition, for cross-scene unsupervised domain adaptation HSI classification, Feng et al. [23] designed a novel shift-sensitive spatial-spectral disentangling learning (S^4DL) method, which can use labeled source domain data and unlabeled target domain data to learn domain-invariant features, and demonstrated excellent performance on multiple cross-scene HSI datasets. Although single-modal HSI data performs outstandingly in land-cover classification tasks, it is difficult to effectively distinguish targets of interest when different land covers with the same spectral characteristics. Therefore, by introducing multisensor data in RS, e.g., LiDAR [24] to provide elevation information and synthetic aperture radar (SAR) [25] to provide amplitude and phase information, i.e., by using complementary between modalities, designing a joint classification method for multimodal RS data is an important way to further improve the accuracy of land-cover classification.

The joint classification of multimodal RS data generally involves three steps, i.e., modal feature extraction, feature fusion, and classification. In order to solve the problem of multimodal joint classification, many excellent DL methods have been proposed. Xu et al. [26] proposed a two-branch CNN, which uses a two-channel CNN framework to extract the spectral spatial features of HSI, and a CNN with cascaded blocks to extract the elevation features of LiDAR, and then fuses the features of these two CNN branches to classify them, and the experiments show that the use of the two branches can improve the model performance. Cai et al. [27] proposed a multimodal fusion network based on graph attention, where the correlation and complementarity between multisource data are considered in the feature extraction module through parameter sharing, in addition, for the purpose of solving the long-distance dependency problem, the feature fusion module uses graph attention fusion. Tang et al. [28] proposed a multiinformation collaborative fusion network by designing a two-branch CMA fusion transformer to obtain global information and promote learning between multimodal features, and also designed an adaptive mask modulation module to dynamically balance the learning rate of modalities to ensure its effectiveness. Xia et al. [29] designed a ternary modality contrastive learning model. Xu et al. [30] proposed a two-branch dynamic modulation network and proposed a new dynamic multimodal gradient optimization strategy.

B. Mamba

In order to fully extract the global context information, an architecture-transformer [31] that can effectively capture the long-range dependency of sequences has been proposed. Transformer eliminates the position dependency problem of the traditional recurrent neural network (RNN) by introducing a self-attention mechanism, which greatly improves the parallelism and efficiency of the model. Roy et al. [3] introduced the multihead cross-patch attention mechanism, which can integrate other complementary modalities and HSI patch tokens for land-cover classification. Zhao et al. [32] proposed a two-branch approach consisting of a hierarchical CNN and transformer, and then designed a cross-token attention fusion encoder to fuse the features of the two branches to improve the performance of joint classification.

Although the transformer shows better performance, its attention computation is characterized by square complexity, which undoubtedly poses some challenges in terms of modeling efficiency and memory overheads. In contrast, the recently proposed Mamba [16] built on state-space models (SSMs), an architecture that employs a hardware-aware algorithm, achieves faster inference and near-linear computational complexity, i.e., it is more efficient than the transformer, and shows great potential for long sequence modeling. In recent studies, Mamba has introduced selective SSM for 1-D sequence modeling along specific directions; in addition, Vim [33] and Vmamba [34] have introduced Mamba to the field of 2-D vision tasks, achieving performance and efficiency improvements in image classification tasks with visual SSM (VSSM) mechanism. Liao et al. [17] proposed a multimodal Mamba fusion module that not only better mines the internal features of multimodal data, but also captures intermodal interrelationships to improve the fusion capability. Wang et al. [35] proposed a tracking Mamba. Ye et al. [36] proposed a high-frequency augmented Mamba network.

C. Knowledge Distillation

Knowledge distillation [37] has gained a lot of attention as a model compression technique in the field of deep learning, which is a machine-learning method that transfers knowledge from a powerful teacher model to a compact student model. This idea was first introduced by Hinton et al. [38], showing that "dark knowledge" can be propagated during model training. In recent years, knowledge distillation has achieved convincing performance in model compression [39], [40]. Luo et al. [41] proposed a graph distillation method that integrates multimodal data in the source domain to learn the target domain that lacks modalities. In addition, there are many other knowledge distillation methods, such as dynamic knowledge distillation [42], self-supervised knowledge distillation [43], and cascaded knowledge distillation [44].

In recent years, in order to overcome some limitations in the training process of deep-learning models and make full use of the model's own knowledge, some self-distillation learning frameworks have been proposed, in which the teacher model and the student model share a common network, which can efficiently extract and transfer information and enhance the model's generalization ability. Hou et al. [45] proposed a self-attention distillation method (SAD), which takes the attention graph obtained from deeper layers of the network as a distillation target for lower layers, obtaining substantial improvements. Hu et al. [46] proposed a hierarchical self-distillation feature learning method, in which the prediction of the next block is used as the soft label of the previous layer to extract deeper knowledge. By adopting this method, the hierarchical features extracted by CNN are fully optimized,

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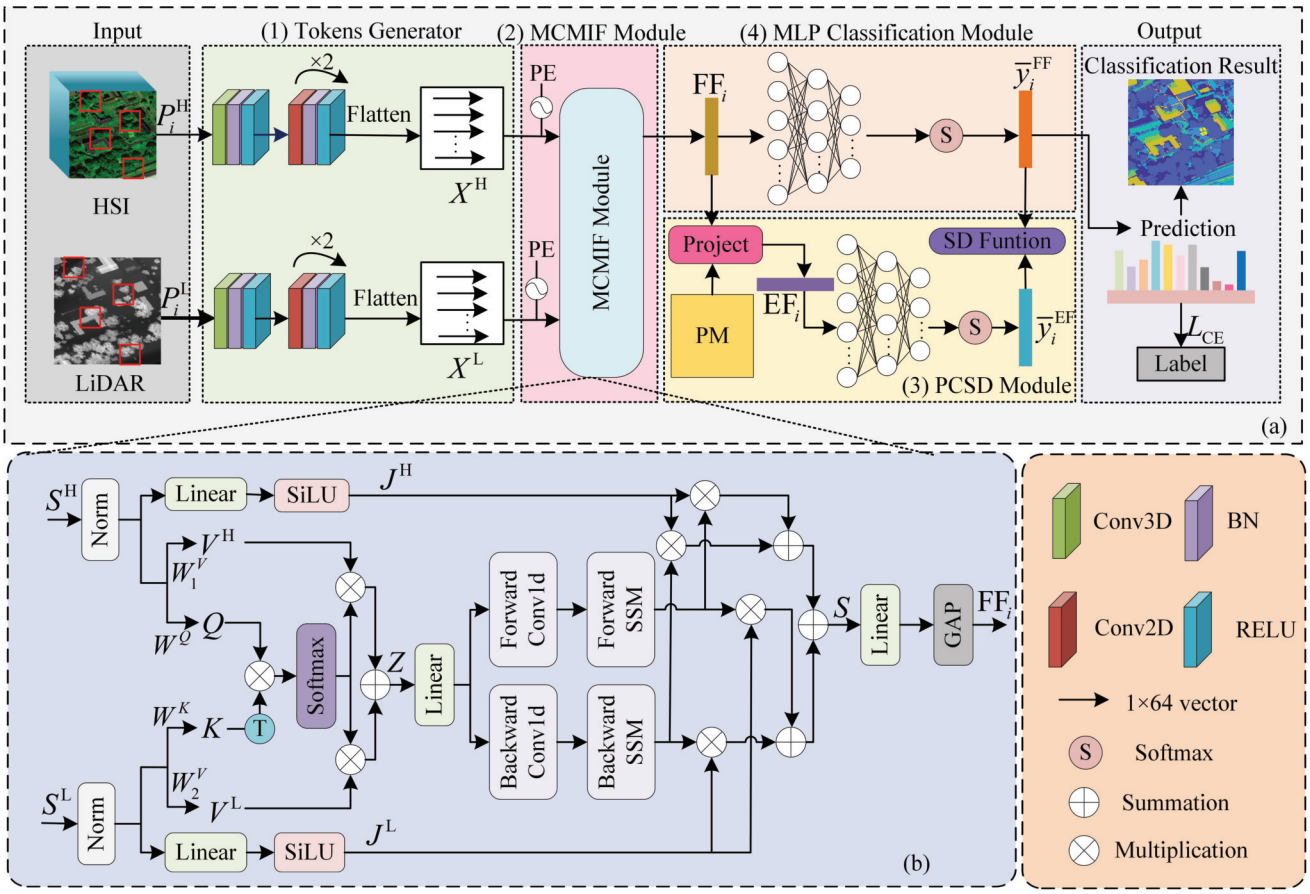


Fig. 1. Mb-CMIFSD. (a) Framework of designed Mb-CMIFSD model. (b) Schematic of the designed MCMIF module.

and the classification accuracy of the model is improved. Sun et al. [47] introduced a hierarchical knowledge distillation approach aimed at transferring knowledge from the dominant modality to other modalities at both the feature and label levels, by establishing intermodal relationships that enhance the performance of nondominant modalities.

III. Designed Mb-CMIFSD Model

In order to further improve the classification accuracy of multimodal RS data by exploiting the complementary information of HSI and LiDAR images, the designed Mb-CMIFSD model is shown in Fig. 1, which mainly contains four modules, i.e., Tokens Generator, MCMIF, PCSD, and MLP Classification modules. The aim is to use Mamba's global context modeling capabilities to capture the complementary information from multimodal RS data that is helpful for joint classification, and combine it with a constructed PCSD knowledge distillation mechanism to further improve the generalization ability of the designed Mb-CMIFSD model.

A. Tokens Generator Design

In the task of joint classification of multimodal RS data, in order to facilitate the subsequent mining of the complementarity between modalities, as shown in *Module* (1) of Fig. 1, we design two different CNN branches for extracting the spectral information of the HSI data and the elevation information of the LiDAR data, respectively. The detailed are as follows.

Let $P_i^{\mathrm{H}} \in R^{s\times s\times B}$ and $P_i^{\mathrm{L}} \in R^{s\times s}$ denote two corresponding patches extracted from the same spatial positions in HSI and LiDAR images, where $s \times s$ denotes the size of the patch, and B denotes the number of spectral bands of HSI. First, for the CNN branch used to extract spectral information from the HSI data, it consists of three convolutional layers stacked sequentially, i.e., the first layer uses eight kernels of size $5 \times 5 \times 9$ for 3-D convolution, while the second and third layers all use HetConv2D convolution [3] with kernel size of 3×3, where the number of kernels is 32 and 64, respectively. Similarly, for the CNN branch used to extract elevation information from LiDAR data, it also consists of three convolutional layers. However, since LiDAR data P_i^{L} is actually a single-channel image, all kernels in this branch are of size 3×3, but the number of kernels in the three layers is 16, 32, and 64, respectively. Notably, to prevent overfitting and accelerate convergence speed during model training, batch normalization (BN) and ReLU activation functions are applied sequentially after each convolutional layer. In summary, for patches P_i^{L} and P_i^{L}, the feature maps obtained by convolving them with corresponding CNN branches are denoted as $F^{\mathrm{H}} \in R^{s\times s\times 64}$ and $F^{\mathrm{L}} \in R^{s\times s\times 64}$, respectively. After flattening, the final tokenization results can be denoted as

$$\begin{cases} X^{\mathrm{H}} = \mathrm{Flatten}\left(F^{\mathrm{H}}\right) \\ X^{\mathrm{L}} = \mathrm{Flatten}\left(F^{\mathrm{L}}\right) \end{cases} \tag{1}$$

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where Flatten(·) denotes the flattening operation along the channel dimension, X^{H} and $X^{\mathrm{L}} \in R^{N\times 64}$ denote the tokenization results corresponding to HSI and LiDAR patches, respectively, $N = s^2$ denotes the length of the sequence.

B. MCMIF Module Design

As shown in (1), after the initial features of multimodal RS data are obtained based on two independent CNN branches, in order to fully exploit the complementarities between multimodal RS data and obtain more discriminative FFs, as shown in Fig. 1(b), a novel MCMIF module is designed, aiming to alleviate the first challenge described in Section I. Specifically, the MCMIF module consists of three key steps, i.e., the CMA mechanism is first employed to realize multimodal semantic alignment. Then, a bidirectional SSM is employed to model the global context information. Finally, a gating mechanism is designed to adaptively perceive the specificity between modalities. In summary, the motivation for designing the MCMIF module is to leverage Mamba's excellent long-term modeling capabilities and its linear computational complexity to exchange information between multimodal RS data and fully explore their complementarity to obtain more discriminative feature representations. The designed MCMIF module is described in detail as follows.

Similar to the ViT [48], the sine and cosine functions are used to calculate the positional embedding $\mathrm{PE} \in R^{N\times C}$ (i.e., $C = 64$), and it is added to X^{H} and X^{L} to obtain the initial input sequences for the designed MCMIF module, i.e.,

$$\begin{cases} S^{\mathrm{H}} = X^{\mathrm{H}} \oplus \mathrm{PE} \\ S^{\mathrm{L}} = X^{\mathrm{L}} \oplus \mathrm{PE} \end{cases} \tag{2}$$

where $\oplus$ denotes element-wise addition, S^{H} and $S^{\mathrm{L}} \in R^{N\times C}$ denote initial sequence with positional embedding information corresponding to the HSI and LiDAR data, respectively.

First, in order to achieve semantic alignment between HSI and LiDAR multimodal RS data, the CMA mechanism shown in (3) is designed to perform the first fusion processing of S^{H} and S^{L}, i.e,

$$\begin{aligned} \mathrm{CMA}\left(S^{\mathrm{H}}, S^{\mathrm{L}}\right) &= \left[V^{\mathrm{H}}, Q, K, V^{\mathrm{L}}\right] \\ &= \left[S^{\mathrm{H}}W_1^{\mathrm{V}}, S^{\mathrm{H}}W^{\mathrm{Q}}, S^{\mathrm{L}}W^{\mathrm{K}}, S^{\mathrm{L}}W_2^{\mathrm{V}}\right] \end{aligned} \tag{3}$$

where W^Q, W^K, W_1^V, and $W_2^V \in R^{C\times C'}$ denote the learnable projection matrices corresponding to Q, K, V^{H}, and V^{L}, respectively, C denotes the dimensions of the input feature. In order to fuse S^{H} and S^{L} into a sequence $Z = \{\mathrm{z}_n | n = 1, 2, \ldots, N\}$, we follow the CMA attention mechanism; the calculation method for each output element $z_n \in R^{1\times C'}$ can be formulated as:

$$a_{n,m} = \exp\left(\frac{1}{\sqrt{C}}\left(Q_n K_m^{\mathrm{T}}\right)\right) \Big/ \sum_{j=1}^{N} \exp\left(\frac{1}{\sqrt{C}}\left(Q_n K_j^{\mathrm{T}}\right)\right) \tag{4}$$

$$z_n = \sum_{m=1}^{N} a_{n,m}\left(V_m^{\mathrm{H}} + V_m^{\mathrm{L}}\right) \tag{5}$$

where $a_{n,m}$ denotes the designed CMA weight calculated using scaled dot products and SoftMax, similar to the cross-attention mechanism.

Second, bidirectional SSM global contextual information modeling. As shown in the middle branch of Fig. 1(b), the $Z \in R^{N\times C}$ is first corrected by layer normalization (LN) processing, and then the dimension is extended to $2C$ by a linear layer to obtain $Z' \in R^{N\times 2C}$, i.e.,

$$Z' = \mathrm{Linear}\left(\mathrm{LN}\left(Z\right)\right) \tag{6}$$

where LN(·) and Linear(·) denote the LN and linear layer projection functions. Next, for sequence $Z' \in R^{N\times 2C}$, it is processed in both forward and backward scanning directions. In each scanning direction, $Z' \in R^{N\times 2C}$ is processed using a 1-D convolutional and a SiLU activation function, then y_{forward} and $y_{\mathrm{backward}} \in R^{N\times 2C}$ are obtained through the SSM layer, i.e.,

$$y_{\mathrm{forward}} = \mathrm{SSM}\left(Z'_{\mathrm{forward}}\right) = \mathrm{SiLU}\left(\mathrm{Conv1d}\left(Z'\right)\right) \tag{7}$$

$$y_{\mathrm{backward}} = \mathrm{SSM}\left(Z'_{\mathrm{backward}}\right) = \mathrm{SiLU}\left(\mathrm{Conv1d}\left(Z'\right)\right) \tag{8}$$

where SSM(·), Conv1d(·), and SiLU(·) represent SSM layer, 1-D convolution, and SiLU activation function, respectively.

Third, an adaptive gating mechanism. As shown in Fig. 1(b), in the top and bottom branches, the input sequences S^{H} and S^{L} are first processed using LN, then the dimensions of the sequences were extended to $2C$ by a linear projection layer, after which they were subjected to the SiLU activation function, respectively, to obtain the adaptive gating activation factors J^{H} and J^{L}, i.e.,

$$\begin{cases} J^{\mathrm{H}} = \mathrm{SiLU}\left(\mathrm{Linear}\left(\mathrm{LN}\left(S^{\mathrm{H}}\right)\right)\right) \\ J^{\mathrm{L}} = \mathrm{SiLU}\left(\mathrm{Linear}\left(\mathrm{LN}\left(S^{\mathrm{L}}\right)\right)\right) \end{cases} \tag{9}$$

where LN(·), Linear(·), and SiLU(·) denote LN, linear layer, and SiLU activation function, respectively.

Finally, in order to use the Mamba mechanism to achieve cross-modal feature fusion, the features y_{forward} and y_{backward} from the forward and backward directions are added and multiplied with the gating factors J^{H} and J^{L} from the output of the top and bottom branches, respectively, to obtain the output sequence $S \in R^{N\times 2C}$. Immediately, S is projected back to the original dimension C through a linear layer, and then after global average pooling (GAP) processing, the cross-modal FF $\mathrm{FF}_i \in R^{N\times C}$ for ith patches P_i^{H} and P_i^{L} is obtained. This process can be formulated as

$$\begin{cases} S = J^{\mathrm{H}} \otimes \left(y_{\mathrm{forward}} + y_{\mathrm{backward}}\right) + J^{\mathrm{L}} \\ \qquad \otimes \left(y_{\mathrm{forward}} + y_{\mathrm{backward}}\right) \\ \mathrm{FF}_i = \mathrm{GAP}\left(\mathrm{Linear}\left(S\right)\right) \end{cases} \tag{10}$$

where $\otimes$ represents element-wise multiplication operation, Linear(·) and GAP(·) denotes the linear project layer and GAP processing function, respectively.

C. PCSD Module Design

As shown in (10), the cross-modal FF FF_i obtained by MCMIF module although has explored the complementarity between multimodal RS data effectively, in order to further enhance the generalization ability of the designed model, as shown in Fig. 2, a novel PCSD module is designed in this section, aiming to improve the discriminative ability of the FFs while further improving the joint classification accuracy of the multimodal RS data.

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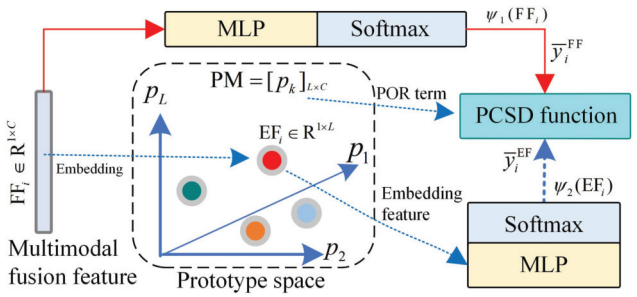


Fig. 2. Schematic of the designed PCSD module.

1) Prototype Matrix and Embedding Feature: In the multimodal RS data joint classification task, although different pixels have the same semantics, their corresponding patches may exhibit different local visual properties due to the influence of their neighboring pixels. Therefore, as shown in Fig. 2, the designed PCSD module first assumes that there are some learnable "*prototype vectors*" in the semantic space, which correspond to the diverse visual properties in the multimodal RS images, i.e., the PCSD module is designed to use the cross-modal FFs of the training data to automatically learn a "*prototype matrix*" PM $= [p_k]_{L\times C}$, where $p_k \in R^{1\times C}$ denotes kth prototype vector, L denotes the number of prototype vectors. Subsequently, for any cross-modal FF FF_i, its projection embedding feature based on the prototype matrix PM is defined as

$$\begin{cases} \mathrm{EF}_i = [u_k : k = 1, 2, \ldots, L] \\ u_k = \dfrac{\langle \mathrm{FF}_i, p_k \rangle}{\|\mathrm{FF}_i\|_2 \cdot \|p_k\|_2} \end{cases} \tag{11}$$

where $\mathrm{EF}_i \in R^{1\times L}$ denotes the project embedding feature, and u_k is actually the cosine distance between FF_i and the kth prototype vector p_k.

2) PCSD Function Design: Let $D = \{(P_i^{\mathrm{H}}, P_i^{\mathrm{L}}, y_i)\}_{i=1}^{\mathrm{BS}}$ represent a batch of multimodal training data, where BS is the batch size, $P_i^{\mathrm{H}}, P_i^{\mathrm{L}}$ and y_i represent the ith training RS data and its category label, respectively. In order to make the cross-modal FFs shown in (10) more excellent and with strong semantic discrimination ability, we improved the standard soft-label knowledge distillation method by introducing prototype constraints and designed a novel PCSD function. Specifically, for each pair of training sample $(P_i^{\mathrm{H}}, P_i^{\mathrm{L}})$ in D, using the methods shown in (1), (10), and (11) in turn, the corresponding cross-modal FFs FF_i and prototype embedding features EF_i can be obtained, which are fed into two different classifiers, and the predicted probability label vectors are denoted as

$$\begin{cases} \bar{y}_i^{\mathrm{FF}} = \psi_1(\mathrm{FF}_i) \\ \bar{y}_i^{\mathrm{EF}} = \psi_2(\mathrm{EF}_i) \end{cases} \tag{12}$$

where $\bar{y}_i^{\mathrm{FF}}$ and $\bar{y}_i^{\mathrm{EF}} \in R^{N_c}$ are the predicted results over N_c categories, respectively, $\psi_1(\cdot)$ and $\psi_2(\cdot)$ are two different two-layer fully connected classifiers with Softmax function.

In the traditional soft-label knowledge distillation, $\bar{y}_i^{\mathrm{FF}}$ is usually used as supervised information to guide the training of $\bar{y}_i^{\mathrm{EF}}$ based on the KL divergence loss [49], i.e.,

$$\begin{aligned} L_{\mathrm{KL}}(D) &= \sum_{i=1}^{\mathrm{BS}} \bar{y}_i^{\mathrm{FF}} \log\left(\bar{y}_i^{\mathrm{FF}} / \bar{y}_i^{\mathrm{EF}}\right) L_{\mathrm{KL}}(D) \\ &= \sum_{i=1}^{\mathrm{BS}} \bar{y}_i^{\mathrm{FF}} \log\left(\bar{y}_i^{\mathrm{FF}} / \bar{y}_i^{\mathrm{EF}}\right). \end{aligned} \tag{13}$$

However, since $\bar{y}_i^{\mathrm{EF}}$ is predicted based on embedding feature EF_i, which is calculated relies on the "*prototype matrix*" PM $= [p_k]_{L\times C}$, where different row vectors (i.e., prototype vectors) represent the visual content corresponding to different local patches of multimodal RS images. If all prototype vectors are uncorrelated, the embedding features EF_i obtained based on PM not only have less redundancy, but also with stronger discriminative ability. Therefore, the cosine distance between any two prototype vectors in PM is computed to construct a *mutual correlation matrix* $\Re \in R^{L\times L}$, and uses it to design a POR term $\|\Re - I\|_2$ for the original KL loss, aiming to constructing a novel PCSD function, i.e.,

$$L_{\mathrm{PCSD}}(D) = L_{\mathrm{KL}}(D) + \lambda \cdot \|\Re - I\|_2 \tag{14}$$

$$\Re = \left[r_{i,j} = \frac{\langle p_i, p_j \rangle}{\|p_i\|_2 \cdot \|p_j\|_2} \right]_{L\times L} \tag{15}$$

where $r_{i,j}$ denotes the cosine distance between the ith prototype vector p_i and the jth prototype vector p_j, $I \in R^{L\times L}$ denotes the identity matrix, $\|\cdot\|_2$ denotes $L2$ norm, and λ is a regulating factor. To summarize, the PCSD function shown in (14) perfectly combines the prototype learning and knowledge distillation techniques, aiming at refining the FF and alleviating the "semantic gap" problem described in the second challenge, and the mechanism of the two components working together is as follows: ① the POR term in the PCSD loss drives the off-diagonal values of $\Re$ toward 0, forcing all prototype vectors to be orthogonal to each other (i.e., cosine distance is 0), making the prototype vectors distributed in different regions of the semantic space and become more diverse and ② the KL term in the PCSD loss aims to use the prediction result of FF to guide the prediction result of embedded feature (EF). In order to improve the accuracy based on EF, it forces the FF of each class of RS data to converge on its exclusive prototype vector and far away from other prototype vectors, making FF have the properties of "intraclass clustering" and "interclass dispersion" in the semantic space, thereby alleviating the problem of "semantic gap."

D. Loss Function and Algorithm Steps

Let $D = \{(P_i^{\mathrm{H}}, P_i^{\mathrm{L}}, y_i)\}_{i=1}^{\mathrm{BS}}$ represent any batch of multimodal training data, where BS represents the batch size, $P_i^{\mathrm{H}}, P_i^{\mathrm{L}}$ and y_i represent the ith training data and its category label, respectively. In order to achieve end-to-end training of the model, based on the L_{PCSD} knowledge distillation function described in (14), and combined it with the standard cross-entropy (CE) classification loss $L_{\mathrm{CE}}(D)$, the total loss function L_{Tot} is defined as follows:

$$\begin{cases} L_{\mathrm{Tot}}(D) = L_{\mathrm{CE}}(D) + L_{\mathrm{PCSD}}(D) \\ L_{\mathrm{CE}}(D) = \displaystyle\sum_{i=1}^{\mathrm{BS}} -y_i \log(\bar{y}_i) \end{cases} \tag{16}$$

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Algorithm 1 Training and Testing of the Mb-CMIFSD Model

Input: Multimodal training dataset TRN = $\{(P_n^{\mathrm{H}}, P_n^{\mathrm{L}}, y_n)\}_{n=1}^{N}$, hyperparameters (Epoches, BS and lr), Test images TST = $\{(P_m^{\mathrm{H}}, P_m^{\mathrm{L}}, y_m)\}_{m=1}^{M}$.

Output: Mb-CMIFSD model Ω and classification accuracy OA.

BS, and randomly initialize the prototype matrix PM using a Gaussian function.

For Epoch in Epoches:

1) Each batch of training data $D = \{(P_i^{\mathrm{H}}, P_i^{\mathrm{L}}, y_i)\}_{i=1}^{\mathrm{BS}}$ is sequentially fed into the tokens generator, MCMIF [i.e., (10)], and prototype embedding [i.e., (11)], we can obtain the cross-modal FFs $\{\mathrm{FF}_i\}_{i=1}^{\mathrm{BS}}$ and prototype embedding features $\{\mathrm{EF}_i\}_{i=1}^{\mathrm{BS}}$.
2) Using the method shown in (12), we can obtain the predicted probability label vectors $\{\bar{y}_i^{\mathrm{FF}}\}_{i=1}^{\mathrm{BS}}$ and $\{\bar{y}_i^{\mathrm{EF}}\}_{i=1}^{\mathrm{BS}}$.
3) Calculate loss using (16), and the optimizer updates the model parameters Ω through backpropagation.
4) Update the learning rate lr using cosine decay strategy.

End for

5) $\forall(P_m^{\mathrm{H}}, P_m^{\mathrm{L}}, y_m) \in$ TST input to successfully trained model Ω, we can obtain $\bar{y}_m$ only using the prediction results corresponding to cross-modal FF.
6) According to $\{(\bar{y}_m, y_m) : m = 1, 2, \ldots, M\}$, statistical the classification accuracy OA and output it.

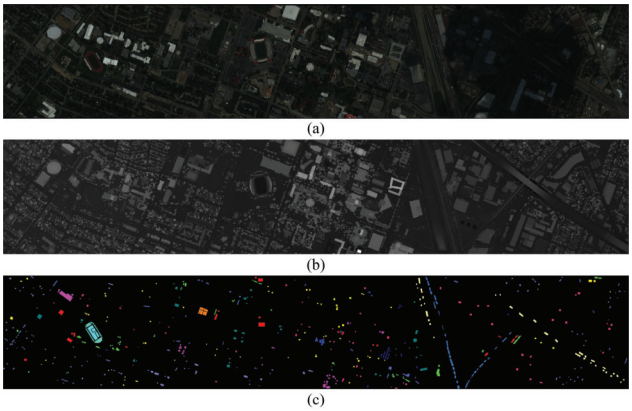


Fig. 3. Visualization of the Houston2013 dataset. (a) Pseudo color image of HSI data using bands 59, 40, and 23, respectively. (b) Gray-scale image of LiDAR data. (c) Ground truth map.

where y_i and $\bar{y}_i \in R^{N_c}$ represent the true label and predicted label of the ith multimodal data, respectively. Finally, in order to better understand the workflow of the designed Mb-CMIFSD model in this article, the network training and testing steps are summarized in Algorithm 1.

IV. Experimental Results and Analysis

A. Datasets

To verify the effectiveness of the designed Mb-CMIFSD model, we conducted comparative experiments on three multimodal datasets, i.e., Houston [50], Trento [51], and MUUFL [52]. The detailed information of resolution, spectral number, wavelength range, and number of annotated pixels for these three datasets is shown in Table I, and Figs. 3–5 depict the pseudo color images of HSI, grayscale images of LiDAR, and ground truth maps of these datasets, respectively. The category names, the corresponding pseudo colors, and the number of training and testing samples for each category included in the three datasets are displayed in the Tables corresponding to Section IV-E.

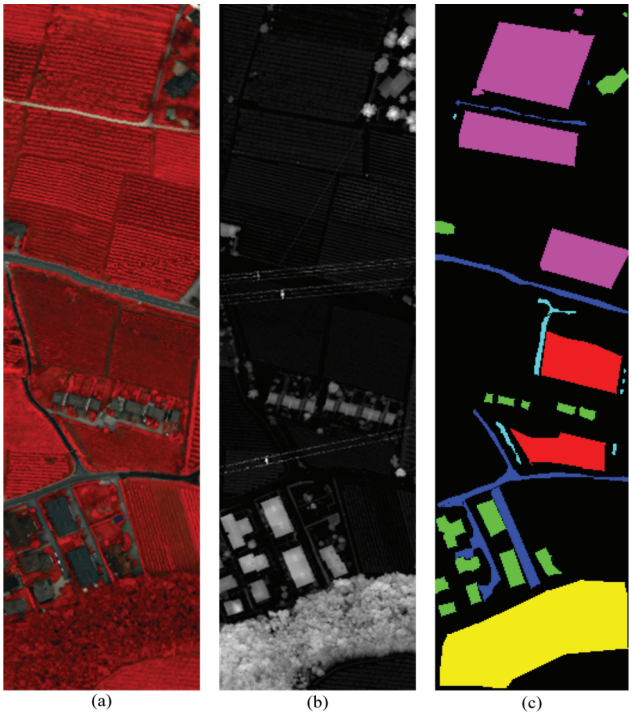


Fig. 4. Visualization of the Trento dataset. (a) Pseudo color image of HSI data using bands 40, 20, and 10, respectively. (b) Gray-scale image of LiDAR data. (c) Ground truth map.

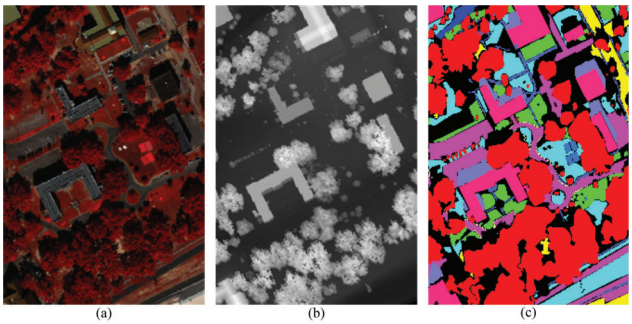
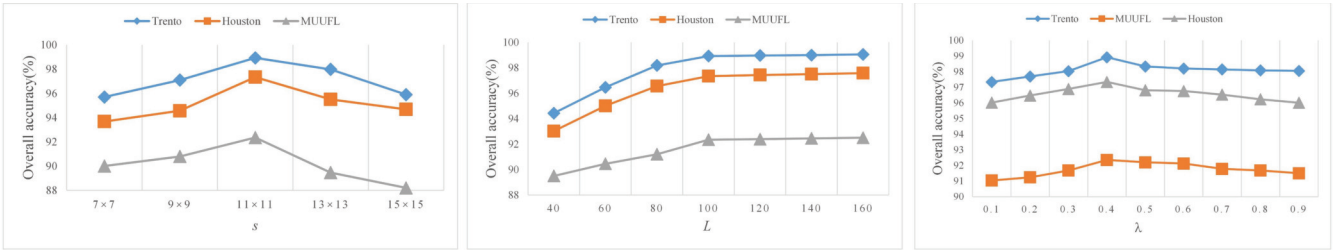


Fig. 5. Visualization of MUUFL dataset. (a) Pseudo color image of HSI data using bands 40, 20, and 10, respectively. (b) Gray-scale image of LiDAR data. (c) Ground truth map.

TABLE I
Specific Details of the Three Datasets

Dataset	Houston2013		Trento		MUUFL	
Class	15		6		11	
Pixels	349×1905		600×166		325×220	
Bands	HSI	144	HSI	63	HSI	64
	LiDAR	1	LiDAR	1	LiDAR	2
HSI wavelength	0.38-1.05μm		0.42-0.99μm		0.38-1.05μm	
Number of labels	15029		30214		53687	

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(a) (b) (c)

Fig. 6. Results of parameter sensitivity experiments based on three datasets for OA. (a) s. (b) L. (c) λ.

B. Experiments Settings and Evaluation Indicators

The proposed model is done on the Pytorch framework, GPU V100-32GB, programming language Python 3.8, torch version 2.0.0. The batch sizes of training and testing were set to 64 and 500. We used the Adam optimizer [53], [54] to train our model for 100 epochs, the learning rate was set as $5e^{-4}$, the weight decay as $5e^{-3}$.

In order to obtain reliable experimental results, we use the overall accuracy (OA), average accuracy (AA), and Kappa (κ) coefficients as evaluation criteria to quantitatively analyze the overall performance of the model, where OA represents the proportion of correctly classified test samples to all test samples, and AA represents the average of class accuracy. The Kappa values reflect the degree of agreement between the classification graphs generated by the proposed model and the truth values provided [27].

C. Parameter Sensitivity Experiments

In the designed Mb-CMIFSD model, there are three key parameters that need to be preset, in order to verify their impact on the model's performance. Based on the Houston, Trento, and MUUFL datasets with OA evaluation index, the following parameter sensitivity experiments are carried out to obtain their optimal values.

1) Patch Size $s \times s$: When generating tokens as shown in (1), in order to verify the impact of patch size on the model's performance, when s is varied from 7 to 15 with a step size of 2, the corresponding OA value is shown in Fig. 6(a). Obviously, the classification accuracy is optimal when s is 11. This is because when s is too small, then the local information that helps classification is not sufficiently included in the patch, thereby it affects the classification accuracy. Conversely, if s is too large, then more neighboring pixels are included in the patch, i.e., more interfering information that is not related to the center pixel's semantics is introduced, which also leads to the degradation of classification accuracy.

2) Number of Prototype Vectors L: In the projection embedding based on the prototype matrix PM shown in (11), in order to verify the impact of the number L of prototype vectors on the performance of the model, when L is varied from 40 to 160 with a step size of 20, the corresponding OA values are shown in Fig. 6(b). Obviously, when L changes from 40 to 100, OA rapidly increases, while L changes from 100 to 160, OA hardly improves further. This is because each prototype vector essentially corresponds to different local features of the RS image, and if L is too small, the visual features corresponding to different semantics cannot be identified, which affects the representation ability of the EFs. While L is too large, which leads to an increase of the EF dimensionality, which increases the amount of data and redundancy, and does not help to further improve the classification accuracy. Therefore, in the subsequent comparative experiments, L was set to 100 to achieve a balance between classification accuracy and data volume.

3) PCSD Regularization Factor λ: In the PCSD loss function shown in (14), in order to verify the effect of the regularization factor λ corresponding to the POR term on the performance of the model, when λ changes from 0.1 to 0.9 with a step size of 0.1, the corresponding OA value is shown in Fig. 6(c). Obviously, the classification accuracy is optimal when λ is 0.4; this is because when λ is too small, the role of the POR term in PCSD is not reflected, while λ is too large, then the POR will be overemphasized and also affect the KL-based information transfer, and the classification accuracy is affected. Therefore, the value of λ was carefully balanced and set to 0.4 in subsequent experiments.

TABLE II
ABLATION EXPERIMENT RESULTS ON THE TRENTO DATASET (%)

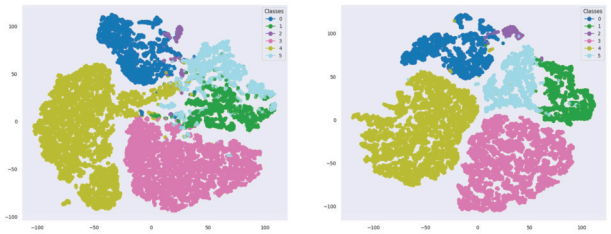
Setting	MCMIF	KL	PCSD	OA (%)	Params(M)
baseline	×	×	×	90.07	0.6137
1	√	×	×	94.36	0.7320
2	√	√	×	96.49	1.0180
Mb-CMIFSD	√	×	√	98.92	1.0242

D. Ablation Experiments

In the Mb-CMIFSD model, to verify the effectiveness of the two key modules designed (i.e., MCMIF and PCSD) in joint classification of multimodal RS data, based on the Trento dataset, the ablation experimental results are shown in Table II. If MCMIF module is not selected, it means that only CMA mechanism [i.e., (3)] is used to fuse multimodal data. Similarly, if KL and PCSD are not selected at the same time, it implies that only standard CE loss is used for model training. Choosing KL means original KL loss is used for self-distillation and combined with the CE loss, while choosing PCSD means POR regularization constraint is implemented on the basis of KL.

As shown in Table II, compared with the baseline model, OA increased by 4.29% when only the MCMIF module is introduced (i.e., Setting 1). Similarly, by introducing KL

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(a) (b)

Fig. 7. Distribution of features obtained with t-SNE algorithm. (a) Baseline. (b) Mb-CMIFSD (ours).

self-distillation (i.e., Setting 2) on the basis of Setting 1, OA further increased by 6.42%. Finally, when both modules are introduced simultaneously, OA reaches the maximum of 98.92%. This indicates that the designed MCMIF and PCSD modules are effective in the joint classification task of multimodal RS data. By observing and analyzing the classification results, we found that compared to the CMA mechanism, the designed MCMIF module combines CMA with the bidirectional Mamba mechanism, which successfully uses Mamba's excellent long-range modeling ability to mine the complementarity of multimodal RS data, thereby obtaining more discriminative FFs and improving the classification accuracy. Meanwhile, the designed PCSD module successfully introduces the prototype learning into the knowledge distillation framework, which is used to refine the FFs during model training, further enhancing the robustness and adaptive ability of the model.

Additionally, in order to visually demonstrate the effectiveness of MCMIF and PCSD, as shown in Fig. 7, we use the T-SNE method [55] to visualize the FFs obtained by the two configuration methods (i.e., baseline and Mb-CMIFSD) in ablation experiments. From the visualization results, it is evident that there are many outliers in Fig. 7(a). However, after being refined by MCMIF and PCSD, the FFs corresponding to each class of samples have better clustering [i.e., Fig. 7(b)], which indicates that the designed MCMIF and PCSD have a positive effect on improving the distinguishability of the features obtained by the model.

E. Comprehensive Comparative Experiments

To demonstrate the effectiveness of the designed Mb-CMIFSD model for joint classification of multimodal RS data, we conducted extensive experiments and compared it with other state-of-the-art methods, including TBCNN [26], CCR-Net [56], EndNet [7], NNCNet [57], CALC [58], MFT [3], Cross [59], and GAMF [27].

1) Results on Houston2013 Dataset: Table III depicts the numerical results of the different methods on the Houston2013 dataset. For the sake of clarity, the best results are denoted in bold, and we can find that Mb-CMIFSD achieves the best performance on OA/AA/Kappa. For example, compared with the best performance model in CNN networks, the designed Mb-CMIFSD model improves 11.12%/11.94%/12.04% on OA/AA/Kappa, respectively, and compared with the best performance model in transformer networks, it improves OA/AA/Kappa by 4.76%/4.03%/5.13%, respectively. From Table III, it can be concluded that our designed model achieves 100% accuracy in the "Soil" and "Tennis Court" categories. Moreover, eight of the 15 categories achieved the highest performance. The excellent performance of the Mb-CMIFSD model is mainly attributed to two reasons. First, the designed MCMIF module can comprehensively interact and mine the complementary information of multimodal RS data, compensating for the differences between modalities and obtaining more discriminative FFs. Second, the designed PCSD module can successfully refine the cross-modal FFs to enhance the robustness and adaptability of feature extraction, while other models without a specifically designed FF refinement mechanism are constrained by the "semantic gap problem," which affects the classification accuracy.

The visualization result of Mb-CMIFSD is shown in Fig. 8(d). Here, we choose to have the best OA methods from the CNN network and the transformer network to compare with the classification results of the designed model, respectively. It is obvious that the classification map of Mb-CMIFSD is the closest to the ground truth, while other algorithms tend to produce more misclassification results and have more noise. For example, in Fig. 8(d), the model we designed achieves clearer categorization with minimal noise in the "Highway" category. At the same time, we can see from Table III that this category indeed achieves the highest accuracy.

2) Results on Trento Dataset: Table IV depicts the numerical results of the different methods on the Trento dataset, and the best results are also denoted in bold in each row. Obviously, compared to all other methods, the designed Mb-CMIFSD model based on OA/AA/Kappa three indicators has the best performance. Specifically, compared to the optimal method in CNN, OA improved by 1.52% (over CCR-Net), AA improved by 3.17% (over TBCNN), and Kappa improved by 2.02% (over CCR-Net). Similarly, compared to the optimal method in transformer, OA improved by 0.2% (over MFT), AA improved by 0.18% (over CALC), and Kappa improved by 0.26% (over MFT). Observing the experimental results shown in Table IV, although designed Mb-CMIFSD model achieves the highest classification accuracy, the improvement based on the Trento dataset was relatively small compared to the Houston 2013 dataset. The main reason is that the Trento dataset only contains six semantics, making it easier to classify. So, all methods have achieved very high classification accuracy (i.e., almost all of them are above 95%), and even a slight improvement is very difficult. Fortunately, the classification accuracy of the designed Mb-CMIFSD model is still improving compared to the other models, which validates the effectiveness of the two designed modules.

In addition, Fig. 9 also provides pixel-level classification maps for various methods. Although the quality of the classification maps of the best CNN-based or transformer-based method is acceptable, the results obtained by Mb-CMIFSD are still the best. For example, in the classification results of the "Buildings" class, Mb-CMIFSD predicts fewer misclassified pixels and smoother boundaries than others. Overall, the prediction map of Mb-CMIFSD is closest to the ground truth, which further confirmed its effectiveness and competitiveness in the joint classification of multimodal RS data.

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TABLE III

OA, AA, κ, AND ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON HOUSTON2013 DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

No.	Class (Train/Test)	CNN Networks			Transformer Networks					Proposed
		TBCNN [26]	CCR-Net [56]	EndNet [7]	NNCNet [57]	CALC [58]	MFT [3]	GAMF [27]	Cross [59]	
1	Healthy grass (20/1231)	90.01	95.94	91.36	78.00	85.41	93.41	87.40	**97.51**	96.87
2	Stressed grass (20/1234)	92.78	92.95	86.47	96.86	93.65	94.89	**98.78**	96.18	97.84
3	Synthetic grass (20/677)	99.94	99.70	96.75	99.71	93.45	**100**	**100**	99.56	99.60
4	Trees (20/1224)	97.76	95.18	88.69	87.12	89.63	91.91	93.79	93.82	**98.77**
5	Soil (20/1222)	97.52	99.18	91.19	99.11	**100**	95.58	99.67	**100**	**100**
6	Water (20/305)	91.41	78.36	93.15	97.14	88.57	**100**	**100**	95.51	97.90
7	Residential (20/1248)	88.13	88.14	59.07	78.22	90.30	81.65	86.69	93.75	**95.90**
8	Commercial (20/1224)	83.53	72.88	57.79	73.58	74.96	90.11	76.14	76.54	**99.62**
9	Road (20/1232)	80.80	62.50	66.94	59.50	79.31	90.82	87.01	79.81	**95.67**
10	Highway (20/1207)	79.99	77.88	69.15	93.18	82.74	92.54	91.79	93.05	**95.27**
11	Railway (20/1215)	84.57	77.61	64.05	**98.69**	98.04	82.96	75.14	89.61	95.54
12	Parking Lot 1 (20/1213)	75.89	79.72	43.31	91.58	92.40	92.74	91.01	93.20	**95.10**
13	Parking Lot 2 (20/449)	7.88	78.40	43.92	96.51	**100**	97.99	93.76	96.97	99.30
14	Tennis Court (20/408)	99.41	92.16	97.55	99.76	**100**	**100**	**100**	99.95	**100**
15	Running Track (20/640)	98.50	95.94	98.61	99.38	**100**	96.25	97.81	99.69	98.31
	OA	86.22	85.36	74.44	87.77	90.03	92.00	90.36	92.58	**97.34**
	AA	84.54	85.77	76.08	89.89	91.23	93.39	91.94	93.68	**97.71**
	Kappa(%)	85.07	84.16	72.87	86.78	89.22	91.36	89.58	91.98	**97.11**

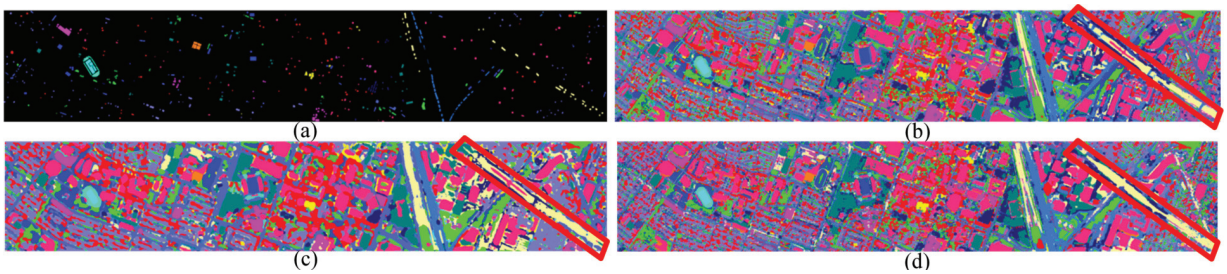


Fig. 8. Classification maps for Houston2013 obtained by different methods. (a) Ground truth. (b) TBCNN. (c) Cross. (d) Mb-CMIFSD (ours).

TABLE IV

OA, AA, κ, AND ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON TRENTO DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

No.	Class (Train/Test)	CNN Networks			Transformer Networks					Proposed
		TBCNN [26]	CCR-Net [56]	EndNet [7]	NNCNet [57]	CALC [58]	MFT [3]	GAMF [27]	Cross [59]	
1	Apple trees (20/4014)	98.74	97.56	89.01	98.21	99.08	97.48	95.26	**99.27**	99.26
2	Buildings (20/2883)	93.42	94.24	92.04	92.78	99.76	98.02	93.09	97.43	**99.89**
3	Ground (20/459)	95.90	80.83	86.41	98.29	**100**	95.42	**100**	99.83	98.40
4	Woods (20/9103)	96.58	99.89	91.09	**100**	**100**	99.89	**100**	99.99	**100.0**
5	Vineyard (20/10481)	93.80	99.14	81.94	92.05	99.04	**99.97**	98.35	99.94	99.80
6	Roads (20/3154)	91.71	89.51	86.48	**97.85**	88.24	93.84	93.27	88.48	89.88
	OA	95.08	97.40	87.16	96.05	98.29	98.72	97.43	98.42	**98.92**
	AA	94.70	93.53	87.83	96.53	97.69	97.44	96.67	97.49	**97.87**
	Kappa(%)	93.80	96.53	83.21	94.77	97.72	98.29	96.58	97.90	**98.55**

3) Results on MUUFL Dataset: The comparative experimental results of different methods on the MUUFL dataset are shown in Table V, and the best results are also highlighted in bold. By observing the experimental data, it can be found that the three metrics OA/AA/Kappa of the designed Mb-CMIFSD model reached the maximum values, i.e., 92.35%, 86.22%, and 89.85%. Moreover, the accuracy of Mb-CMIFSD reached the maximum value in six of the eleven categories of the MUUFL dataset. Specifically, taking the best CNN-based or transformer-based method as reference, our Mb-CMIFSD model achieves 13.56% or 8.78% improvement on their Kappa metrics, respectively. In addition, a notable finding is that in "Mixed ground surface," the improvement of Mb-CMIFSD is significant, 13.73% ahead of other methods with the highest accuracy (i.e., 74.10%) in this category. This is because pixels classified as "Mixed ground surface" in RS data correspond to more diverse and complex land covers, which are more obviously interfered with by the "semantic gap." However, in the designed Mb-CMIFSD model, the MCMIF module integrates CMA and a bidirectional Mamba mechanism, which can better use contextual information to explore the complementarity between modalities. Especially, the designed PCSD module successfully integrates prototype learning and knowledge distillation, which can refine

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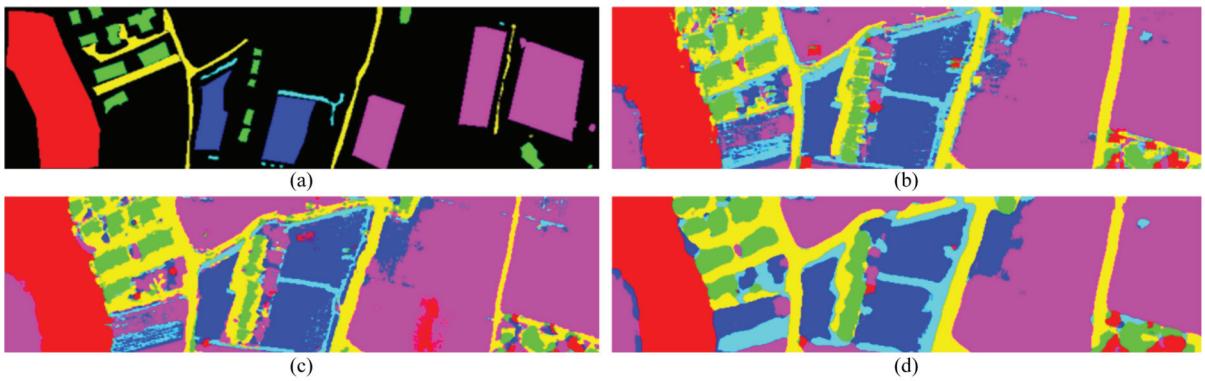


Fig. 9. Classification maps for Houston2013 obtained by different methods. (a) Ground truth. (b) CCR-Net. (c) MFT. (d) Mb-CMIFSD (ours).

TABLE V

OA, AA, κ, AND ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON MUUFL DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

No.	Class (Train/Test)	CNN Networks			Transformer Networks					Proposed
		TBCNN [26]	CCR-Net [56]	EndNet [7]	NNCNet [57]	CALC [58]	MFT [3]	GAMF [27]	Cross [59]	
1	Trees (20/23226)	84.52	76.63	79.70	80.31	86.35	88.94	89.99	91.68	**97.68**
2	Mostly grass (20/4250)	79.64	63.22	74.33	**85.05**	54.48	75.97	75.01	67.60	84.27
3	Mixed ground surface (20/6862)	65.23	74.10	61.31	50.80	51.32	59.80	65.06	71.55	**87.83**
4	Dirt and sand (20/1806)	80.66	74.03	74.11	85.57	90.31	81.00	90.75	89.39	**94.75**
5	Road (20/6667)	82.04	81.51	83.83	65.60	64.46	69.11	88.34	83.70	**93.28**
6	Water (20/446)	97.76	97.76	96.57	98.03	**100**	99.77	99.77	96.23	91.87
7	Building shadow (20/2213)	81.62	71.62	86.16	76.88	84.89	85.04	93.04	89.77	**94.44**
8	Building (20/6220)	94.23	87.57	90.33	81.17	89.50	85.77	94.11	94.23	**96.49**
9	Sidewalk (20/1365)	49.60	**64.03**	60.35	41.82	30.11	45.71	51.57	56.26	43.50
10	Yellow curb (20/163)	70.31	87.12	**91.17**	62.43	54.34	71.77	84.66	73.19	81.15
11	Cloth panels (20/249)	95.58	93.17	95.38	97.30	95.75	95.98	95.58	**97.83**	83.20
	OA	81.45	76.79	78.50	74.39	75.65	79.87	85.14	85.43	**92.35**
	AA	80.11	79.16	81.20	74.99	72.86	78.08	84.36	82.86	**86.22**
	Kappa(%)	76.29	70.80	72.78	67.60	68.85	74.08	80.79	81.07	**89.85**

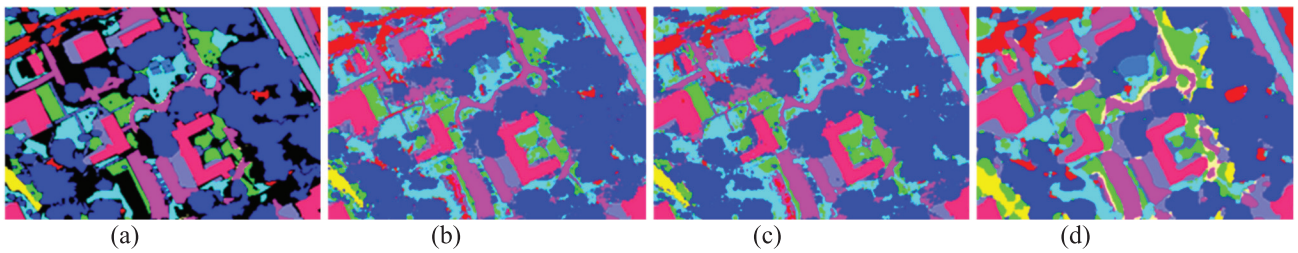


Fig. 10. Classification maps for the Houston2013 obtained by different methods. (a) Ground truth. (b) TBCNN. (c) Cross. (d) Mb-CMIFSD (ours).

cross-modal FFs and enhance the robustness of the feature extraction of the model. These experimental results also confirm that PCSD is effective in alleviating the "semantic gap."

Meanwhile, we also select the best OA methods from CNN and transformer networks, respectively, and the visualized comparative classification results with the designed Mb-CMIFSD model are shown in Fig. 10. By observing the classification results, we find that our method Mb-CMIFSD, can distinguish different categories more accurately and is closest to the ground truth map. Specifically, taking the classification result of the "Road" category as an example, Mb-CMIFSD has the least number of incorrectly predicted pixels, which means that the area classified as "Road" is more complete, and the boundary is also clearer and smoother.

F. Computational Complexity Analysis

In the field of multimodal RS data classification, the computational complexity of the model is a core consideration factor. In general, the two key metrics for assessing model complexity are floating-point operations (FLOPs) and parameters. In view of this, we conducted experiments on different methods using the Houston 2013 dataset. Fig. 11 represents the scatter plot of OA versus parameters and FLOPs. From Fig. 11, it is easy to see that the designed Mb-CMIFSD model exhibits the highest OA value and appropriate computational complexity. Specifically, Mb-CMIFSD is significantly lower than GAMF in terms of the number of parameters and FLOPs, while the classification accuracy is higher than that of GAMF. In addition to this, although Mb-CMIFSD does not have the lowest FLOPs and parameters among all the models, its excellent performance enhancement completely compensates

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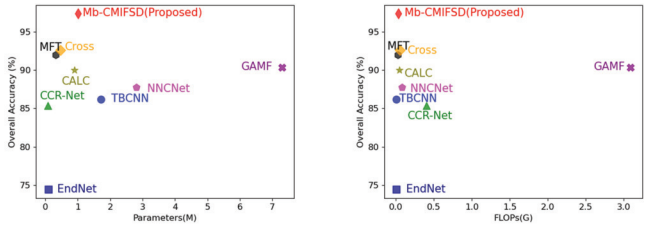


Fig. 11. Computational complexity analysis results. (a) OA versus Parameters and (b) OA versus FLOPs on the Houston 2013 dataset.

for this shortcoming. For example, although the parameters of the Mb-CMIFSD model are more than those of CCR-Net, the OA value is more than 10% higher than the CCR-Net. This suggests that reducing the computational complexity of the model and pursuing a lightweight and efficient model design may lead to a decrease in classification accuracy.

V. Conclusion

In order to fully exploit the complementarity between multimodal RS data and enhance the generalization ability of the model, a new model called Mb-CMIFSD is designed in this article, which contains two main core modules, i.e., MCMIF and PCSD. The ablation experiment results (Table II) show that compared with the baseline model, the introduction of the MCMIF module in Mb-CMIFSD increases OA by 4.29%, and then introducing the PCSD model, OA further increases by 8.85%. In summary, in the designed Mb-CMIFSD model, the MCMIF module successfully integrates CMA with a bidirectional Mamba mechanism, which can use Mamba's excellent long-distance modeling ability to fuse the original features from two modalities. Moreover, the PCSD module perfectly integrates prototype learning and knowledge distillation, and uses the constructed prototype constraints to regularize the KL self-distillation function to further refine the cross-modal FFs, thus enhancing the generalization ability of the model.

Although the designed Mb-CMIFSD model has excellent classification performance and robustness, it still has some shortcomings, i.e., the MCMIF module integrates cross-modal cross-attention with the Mamba mechanism, which is not lightweight enough and has a higher computational cost compared to pure Mamba. Therefore, our future work is to design more efficient lightweight cross-modal data fusion mechanisms, explore new knowledge distillation methods, and further validate the robustness and generalization ability of the designed model in RS datasets with more modalities and larger scales.

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